A Three-Head Multitask Framework for Glioma Segmentation, Mutation Classification, and Survival Analysis

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December 3, 2025

1 Introduction

Glioma is a type of brain cancer that grows in the brain's support tissue. When a patient is diagnosed, a clinician needs to learn several things about the tumor. Three key items are especially important for decision-making: where the tumor is and how far it extends, what its genetic mutations are, and the patient's prognosis. The goal of this work is to estimate these three items automatically and non-invasively using MRI images, since MRI is the standard method used to visualize glioma.

Segmentation is the computational process of converting MRI scans into predicted tumor regions (binary mask) that separate tumor from normal tissue. This process transforms visual imaging data into structured information that can be quantitatively analyzed. By outlining tumor boundaries, segmentation allows estimation of tumor size, volume, and spatial distribution within the brain, though the reliability of these measurements depends on the accuracy of the segmentation. In clinical workflows, segmentation results can assist in treatment planning—for example, determining how much tissue to remove surgically or where to focus radiation therapy. Beyond clinical use, segmentation masks also support research applications, where they serve as inputs for models that aim to predict molecular characteristics or patient outcomes [9, 7].

Recent studies integrate molecular data with imaging to better understand tumor behavior and patient outcomes. Radiogenomics integrates medical imaging and genomics to produce complementary insights—often by identifying associations between MRI-derived features and tumor genomic characteristics [3]. Models for *mutation classification* from MRI aim to infer gene-level alterations—such as IDH1, EGFR, PTEN, and TP53—from image-derived features. The motivation is practical: finding mutation information requires removing brain tissue, which is invasive, while MRI is already routinely acquired. *Survival analysis*, often implemented through models such as the Cox proportional hazards (CoxPH) [14] framework, estimates time-to-event outcomes using imaging features and clinical data [5].

Automated glioma segmentation is a well researched domain. Ronneberger *et al.* [9] introduced U-Net, which popularized the encoder–decoder design with skip connections for biomedical imaging, allowing voxel-level predictions even with limited data. Isensee *et al.* [7] proposed nnU-Net, a self-configuring framework that standardizes preprocessing, architecture selection, and loss design. In brain tumor segmentation, nnU-Net consistently delivers strong out-of-the-box performance and serves as the default baseline for evaluation.

Radiomic features are numerical measurements extracted from the tumor region, such as size and shape. Gutman *et al.* [2] showed that radiomic features are linked to specific genetic mutations in glioma, suggesting that imaging can reveal information about a tumor's genetic makeup and aggressiveness. Kazerooni *et al.* [3] later summarized this evidence, highlighting consistent MRI–genotype associations such as EGFR amplification patterns and core/necrosis signatures linked to TP53 mutations. These findings motivated further exploration of image-based mutation prediction as a potential proxy when molecular testing is incomplete or delayed.

Poursaeed *et al.* [18] reviewed recent machine learning and deep learning methods for survival prediction in glioma, and reported that most models use Cox proportional hazards with image features to estimate survival risk. Heagerty *et al.* [5] developed evaluation measures for survival models that assess whether higher predicted risk corresponds to shorter observed survival times, while correctly handling patients who are still alive at the end of follow-up. Together, these studies show that survival prediction from imaging is active and well-studied, but primarily in a single-task setting.

Even though segmentation, mutation classification, and survival modeling are used together in practice, most predictive models are still trained separately. Multi-task learning aims to train a single model to perform several related tasks simultaneously, allowing shared representations to improve prediction across tasks. There are many challenges with this methodology, which could be the reason behind its slow adoption. The tasks operate at different granularities:

segmentation produces dense, voxel-level predictions, while mutation and survival modeling make patient-level predictions. These differences can lead to conflicts when the tasks share a single encoder. Sener *et al.* [8] framed joint optimization as a multi-objective problem, emphasizing that each task defines its own loss landscape and that naïve averaging can degrade performance for others. To address task-level trade-offs more systematically, Kendall et al. [4] introduced *homoscedastic uncertainty weighting*. This approach learns task weights from their estimated uncertainty so noisier tasks contribute less during training and the model remains more stable.

While segmentation tasks benefit from abundant labeled data, mutation classification has far fewer labeled cases, making it difficult to train reliably. Semi-supervised learning offers a potential way to alleviate this imbalance by incorporating unlabeled subjects through pseudo-labeling, but these methods require explicit uncertainty handling to avoid reinforcing incorrect predictions. Gal and Ghahramani [6] expanded this uncertainty-based approach by showing that Monte Carlo dropout can approximate Bayesian uncertainty, enabling the model to identify and filter unreliable pseudo-labels in semi-supervised settings.

Cheng *et al.* [1] applied these ideas to glioma by jointly training segmentation and mutation classification with uncertainty-aware weighting and pseudo-labeling. They reported improvements over single-task models, but only for one mutation and on a relatively small test set, so it remains unclear whether these gains extend to other genes and to survival prediction. Overall, these studies indicate that combining strong segmentation models with uncertainty-based weighting and semi-supervised learning provides a solid basis for multi-task learning in glioma.

We study a multi-task learning framework that jointly models glioma segmentation, mutation classification, and survival prediction from multimodal MRI images. The network receives four MRI sequences (T1, T1ce, T2, and FLAIR) and shares a common encoder whose output features feed three task-specific heads: one for tumor segmentation, one for per-gene mutation prediction (EGFR, IDH1, PTEN, TP53), and one for survival risk estimation. This design allows spatial information learned during segmentation to support the downstream tasks of mutation classification and survival prediction. Because molecular labels are limited compared to segmentation masks, we integrate an uncertainty-aware semi-supervised strategy for mutation classification that selects reliable pseudo-labels to improve prediction without additional annotation. We compare single-task and multi-task configurations to test whether sharing information across tasks improves performance.

Our contributions are twofold. (i) We develop a unified three-head multi-task model for voxel-wise segmentation, per-gene mutation prediction (EGFR, IDH1, PTEN, TP53), and survival risk estimation from the same multimodal MRI inputs. (ii) We also evaluate a variant that fuses radiomic features extracted from the predicted tumor masks with the learned image features, in order to test whether explicit shape information improves mutation classification and survival analysis.

2 Glioma Segmentation

Glioma segmentation aims to delineate tumor regions from MRI scans, providing a voxel-level representation of disease extent. The task typically separates the tumor into biologically and clinically meaningful components—*enhancing core*, reflecting contrast uptake and blood–brain barrier disruption; *non-enhancing or necrotic core*, corresponding to regions of cell death or reduced perfusion; and *peritumoral edema*, indicating tissue swelling and infiltrative spread in surrounding white matter [12]. We implement a 3D U-Net for automated glioma segmentation, using multimodal MRI inputs to generate detailed voxel-level maps of tumor subregions.

2.1 Dataset

We use the BraTS 2023 Adult Glioma Training Set (`https://www.synapse.org/#!Synapse:syn51156910`) comprising preoperative brain MRI scans with tumor subregion labels [10]. For each subject we include four structural MRI modalities: T1-weighted (T1), contrast-enhanced T1 (T1CE), T2-weighted (T2), and Fluid-Attenuated Inversion Recovery (FLAIR). These modalities provide complementary information—T1 for anatomical detail, T1CE for regions with blood–brain barrier disruption, T2 for fluid-sensitive signal, and FLAIR for edema visualization with cerebrospinal fluid suppression [11]. The dataset is multi-institutional and publicly released with standardized annotations for enhancing tumor, necrotic or non-enhancing core, and peritumoral edema following the BraTS protocol [10]. In our experiments we use the full annotated training cohort (approximately 1,251 adult cases) and establish a fixed patient-wise split that is reused across all experiments.

2.2 Preprocessing

We compute a fixed 3D bounding box for the entire cohort by scanning all subjects and recording the minimum and maximum voxel coordinates that contain any nonzero intensity. This bounding box defines the smallest region encompassing all brains in the dataset. Each case's four MRI modalities are then cropped to this region, reducing memory usage by roughly 30% and allowing the model to focus on relevant tissue. The final bounding box parameters are

$$\text{BBOX} = (z_0, z_1, y_0, y_1, x_0, x_1) = (37,\ 216,\ 10,\ 229,\ 0,\ 155), \tag{1}$$

which correspond to the global minimum and maximum nonzero coordinates computed from the BraTS 2023 training cohort. Within this crop, each modality is z-score normalized on a per-case basis:

$$x' = \frac{x - \mu}{\sigma + 10^{-5}}. \tag{2}$$

In this expression, x is the raw voxel intensity, μ and σ are its percase mean and standard deviation, and x' is the normalized value. Cropped, normalized images are stacked channel-wise in the order `[T1, T1CE, T2, FLAIR]` to form a $(4, D, H, W)$ tensor. Each case is saved as a paired `(image, label)` tensor for consistent, fast loading across tasks.

2.3 Methodology

Training is fully supervised with a 70%/10%/20% patient-level split for train, validation, and test. We train the segmentation network with a soft Dice loss, which directly optimizes overlap between the predicted mask and the ground truth and emphasizes the tumor region itself rather than background accuracy. We use Adam (learning rate 10^{-4}, β_1=0.9, β_2=0.999) with a batch size of 2 for up to 100 epochs, and apply early stopping after seven epochs to limit overfitting.

2.3.1 Network Architecture

A 3D U-Net architecture was implemented, as encoder–decoder networks with skip connections remain a robust baseline for 3D medical image segmentation, particularly in brain tumors [9, 7]. The encoder has four resolution levels followed by a bottleneck; each level applies a `DoubleConv3D` block (two 3×3×3 convolutions with GroupNorm(8) and ReLU). We used GroupNorm over BatchNorm due to small batch sizes (batch = 2), which can make BatchNorm statistics noisy. Downsampling uses 2×2×2 max-pooling, with channels increasing 16→32→64→128; the bottleneck uses 256 channels to aggregate global context.

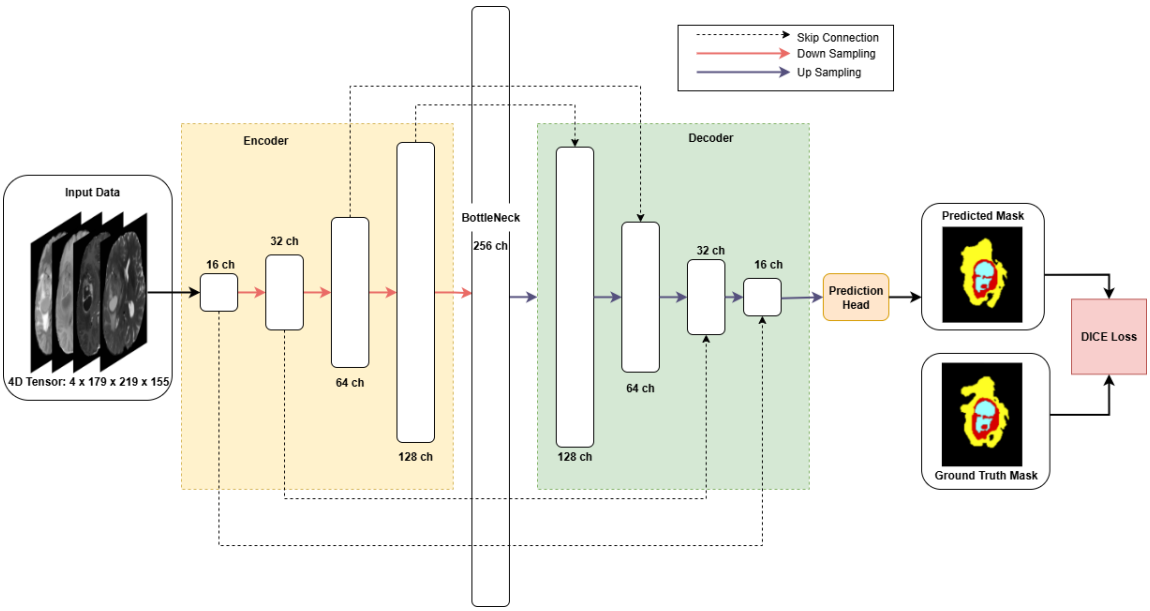

Figure 1: Segmentation pipeline. Multimodal MRI (T1, T1CE, T2, FLAIR) are preprocessed and passed to a 3D U-Net with four encoder–decoder levels and skip connections. A $1\times1\times1$ convolution produces four logits (background and three tumor subregions), and a softmax yields per-voxel probabilities.

The decoder mirrors the encoder and upsamples using transposed convolutions (kernel = 2, stride = 2). At each level, features from the encoder are concatenated through skip connections which are particularly important for small, low-contrast foci (e.g., non-enhancing core), where direct propagation of early high-frequency information improves boundary fidelity [9]. The decoder then applies the same `DoubleConv3D` blocks to integrate high-resolution detail with deeper context, as shown in Fig. 1. Decoder feature widths decrease $128\rightarrow64\rightarrow32\rightarrow16$, and a final $1\times1\times1$ convolution maps to four output logits.

3 Mutation Classification

Accurate classification of gene mutations is essential, as it offers critical insights into tumor biology and supports personalized prognostic and therapeutic decision-making. In our pipeline, this task operates on image representations derived from the segmentation bottleneck and radiomic features extracted from the segmentation masks. The goal is to predict per-gene mutation status from these imaging features, even when labels are limited or partially missing. In addition to point mutations, we also consider copy-number alterations (CNA), including *amplifications* (extra copies) and *deletions* (loss of copies).

3.1 Dataset

Subjects used for mutation/CNA prediction were derived from the subset of cases with both imaging and matching genomic data. In total, 29 subjects were available. The distribution of positive cases was highly imbalanced across genes (IDH1: 4 positives, EGFR: 11 positives, TP53: 9 positives, PTEN: 12 positives). Genomic labels (point mutations and copy-number alterations) were obtained from cBioPortal (https://www.cbioportal.org/) [15]. For each subject, inputs to the classifier consisted of imaging-derived representations produced by the trained segmentation model: (i) bottleneck embeddings exported per subject, and (ii) masks predicted by the segmentation head from which radiomic features were computed. Gene-level labels were then attached per subject for point mutations and CNA.

3.2 Preprocessing

A 1D deep feature vector was constructed per subject from the segmentation outputs. The vector was extracted from the 3D U-Net *bottleneck*. Concretely, we forward the four-channel MRI tensor through the encoder path and stop at the bottleneck, obtaining an activation tensor $B \in \mathbb{R}^{C_b \times D' \times H' \times W'}$, where C_b is the bottleneck channel count (256 in our setup) [16]. We then apply global average pooling over the spatial axes, producing a subject-level embedding $f \in \mathbb{R}^{C_b}$. Each component is computed as

$$f_c = \frac{1}{D'H'W'} \sum_{i=1}^{D'} \sum_{j=1}^{H'} \sum_{k=1}^{W'} B_{c,i,j,k}, \quad c = 1, \ldots, C_b. \tag{3}$$

Here, B is the bottleneck activation tensor, C_b is number of bottleneck channels, and (D', H', W') are the spatial dimensions of the bottleneck feature map.

In parallel, for each case, 289 radiomic features total were extracted from necrotic, edema, enhancing, and whole–tumor masks using the PyRadiomics toolbox (online documentation) [17], and then concatenated into a single feature vector per subject.

The deep and radiomic features were concatenated into a 544-dimensional vector and standardized by z-scoring using training-fold statistics. We then fit three separate PCA transforms (each fit using only training data): (i) a PCA on the 544-D combined vector (reduced to 35 components), (ii) a PCA on deep-only features (6 components), and (iii) a PCA on radiomics-only features (37 components). For each single-head mutation classification task, we evaluated all six feature variants (raw vs. PCA $\times$ deep-only, radiomics-only, combined) and selected the best-performing representation.

3.3 Methodology

The mutation classification pipeline uses a shared MLP trunk with per-gene output heads for four genes (EGFR, IDH1, PTEN, TP53), enabling parameter sharing when labels are sparse [1]. We study two settings: a mutation+CNA regime (to test whether copy-number alteration information improves discrimination) and a mutation-only regime (to isolate point-mutation prediction) [3, 2]. Each per-gene output is a single binary logit (sigmoid). For example, for the gene EGFR in the mutation-only setting the label is simply `MUT_EGFR` $= 0/1$. When CNA is included, each gene instead has three binary targets: `MUT`, `CNA_AMP`, and `CNA_DEL` (e.g., `MUT_EGFR`, `CNA_AMP_EGFR`, `CNA_DEL_EGFR`), each with a value of 0 or 1.

For training, we perform stratified cross-validation. This divides the subjects into non-overlapping subsets (folds) and cycles through them so that every subject is used for both training and validation across the full procedure. Folds are constructed so that each gene has at least one positive and one negative case in every split. We balance the optimization in two ways: per-class weighting up-weights rare positives relative to negatives within each gene, and per-task weighting prevents genes with more available labels from dominating the gradient.

Pseudo-labeling uses model-generated labels so that unlabeled subjects can contribute to training. We apply this because mutation annotations are sparse, and unlabeled MRIs can still help if we only add predictions that are confident and stable. Following the uncertainty-aware approach of Cheng *et al.* [1], we run Monte-Carlo dropout, average the per-gene sigmoid probabilities, and compute their predictive variance. A pseudo-label for a given subject–gene is accepted only if the mean probability is at least 0.9 and the variance is at most 0.05. Accepted pseudo-labels are appended to the training set, and we then run one additional training round within each fold. This mirrors the UPS criterion in Cheng *et al.* [1], applied independently to each gene.

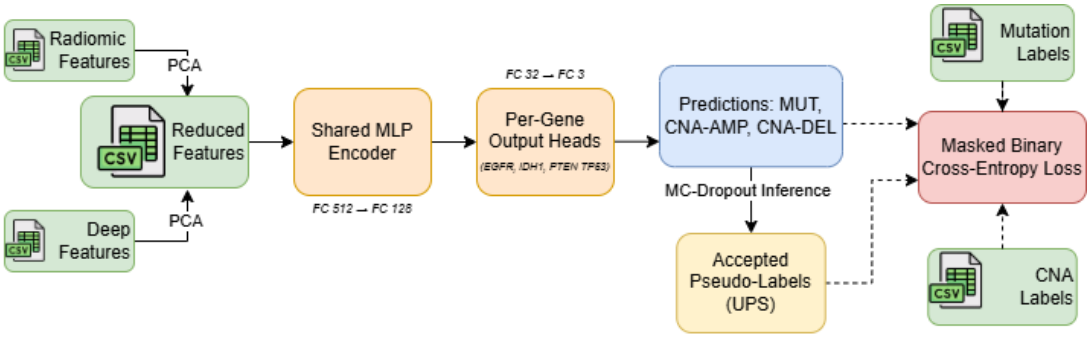

Figure 2: Mutation+CNA setup. The fused feature vector is passed to a shared MLP encoder, and each gene head outputs three binary logits (MUT, CNA-AMP, CNA-DEL). Ground-truth labels from both mutation and CNA supervise the loss; high-confidence pseudo-labels (UPS) are also used.

In the mutation+CNA configuration (Fig. 2), the concatenated radiomics+deep feature vector is optionally PCA-reduced, then passed to a shared multi-layer perceptron encoder. For each gene, a linear head emits three binary logits (MUT, CNA_AMP, CNA_DEL). Unlabeled subjects are added using the same uncertainty-based pseudo-label procedure described above.

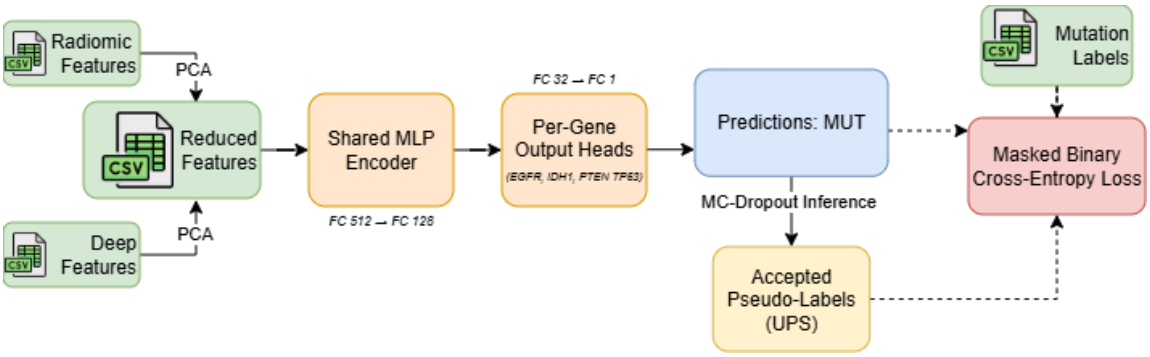

Figure 3: Mutation-only variant. Radiomics+deep features are standardized, optionally PCA-reduced, passed through a shared MLP encoder, and each gene head emits a single binary mutation logit. High-confidence pseudo-labels (dashed path) are appended to training when accepted.

In the mutation-only variant (Fig. 3), CNA branches are removed and each per-gene head emits a single binary mutation logit. The rest of the pipeline remains identical.

4 Survival Analysis

Accurate survival estimation is important. Glioma patients can have dramatically different outcomes; knowing who is likely to survive longer or shorter changes what we actually do. Prognosis drives the level of aggressiveness of surgery and determines prioritization of care [18]. Survival models are also important for research: they allow improvement of patient grouping by risk level, and they allow evaluating whether a new type of therapy improves survival. [19]. In this section, the model receives two types of inputs: (i) image-derived representations (global deep embeddings and mask-based radiomics) and (ii) clinical covariates (age, sex, and therapy). The supervised targets are overall-survival time and an event indicator. Because many subjects are censored (alive at last follow-up), we use a Cox proportional hazards formulation [14], which optimizes a ranking objective that is formally defined to handle this censoring.

4.1 Dataset and Preprocessing

We use the same imaging-derived features as in the mutation section: encoder/bottleneck embeddings from the trained segmentation network and radiomic features computed from predicted masks (see Sec. 3 for feature construction and PCA settings). The survival cohort consists of N=91 subjects also downloaded from cBioPortal, each with overall–survival time (in months) and an event indicator (death=1, censored=0), which are the two supervised targets in this section.

The imaging feature vector is concatenated with clinical covariates. Age is standardized using z-scoring, and sex and therapy are encoded as one-hot variables. Therapy categories in this cohort include Standard Radiation, Nonstandard Radiation, TMZ Chemotherapy, Unspecified Radiation, Unspecified Therapy, Standard Radiation+TMZ Chemotherapy, Nonstandard Radiation+TMZ Chemotherapy, and Standard Radiation+Alkylating Chemotherapy. The resulting tabular vector (imaging PCs plus encoded clinical variables) is used as the input to the Cox model.

4.2 Methodology

We model relative risk using a Cox proportional hazards model (CoxPH) [14]. CoxPH is the classical survival model in biostatistics; it learns how inputs shift the *hazard* of death [5]. CoxPH optimizes a *ranking* loss: it increases the risk score for patients who died earlier and decreases it for patients who survived longer, so that the model is trained to correctly order patients by relative hazard. In CoxPH, the hazard for inputs $x \in \mathbb{R}^d$ is

$$h(t \mid x) = h_0(t) \exp(\beta^\top x), \tag{4}$$

where $h_0(t)$ is the baseline hazard and $\eta = \beta^\top x$ is the linear predictor (risk score the produces the ordering). We feed PCA-compressed deep+radiomics features (and available clinical covariates) to the model. To ensure reproducible and stable validation, the survival experiments use the same fixed patient-wise split as the segmentation task. Model discrimination is evaluated on the held-out validation set using the metrics defined in Sec. 5.5.

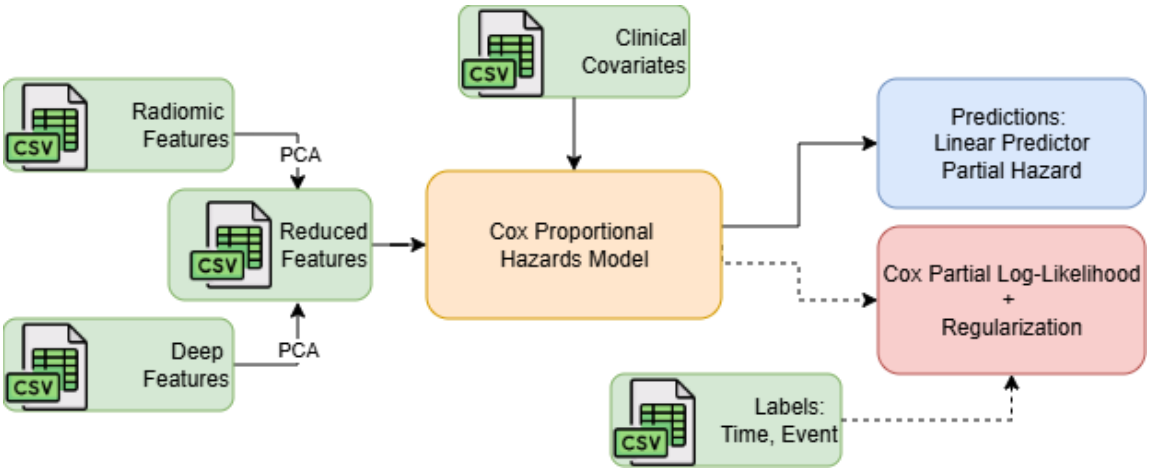

Figure 4: Survival pipeline. 1D image features and clinical covariates are fed to a Cox model that ranks patients by relative risk.

As illustrated in Fig. 4, the Cox proportional hazards model learns coefficients β that assign higher risk scores $\eta = \beta^\top x$ to patients who tend to experience earlier events. Training maximizes the partial log-likelihood, which accounts for censoring by comparing each observed event to all subjects still under observation at that time (the *risk set*) [14]. In practice, this evaluates

whether imaging and clinical features together allow patients to be ranked in the correct order of risk over time.

5 Multi-task Learning Framework

Multi-task learning trains a single encoder to support multiple related tasks. Features learned for segmentation may also be informative for the patient-level tasks of *mutation* classification and *survival* prediction. We test whether sharing one encoder across these three tasks improves performance [8]. Task weights are adjusted automatically during training using *homoscedastic uncertainty weighting* [4]. We also implement a variant that adds radiomic features extracted from the predicted tumor masks to the mutation and survival heads [3] to test whether explicit shape information can improve results.

5.1 Data and Preprocessing

All three heads—segmentation, mutation classification, and survival analysis—use the same subjects, fixed patient splits, and MRI preprocessing described previously.

5.2 Methodology

We attach three task heads to a shared encoder so that segmentation, mutation, and survival prediction are learned jointly from the same features. All training uses the same learning rate and patient-level splits described previously. This design tests whether one encoder can learn from both region-level and patient-level objectives.

5.2.1 Multi-Head Architecture

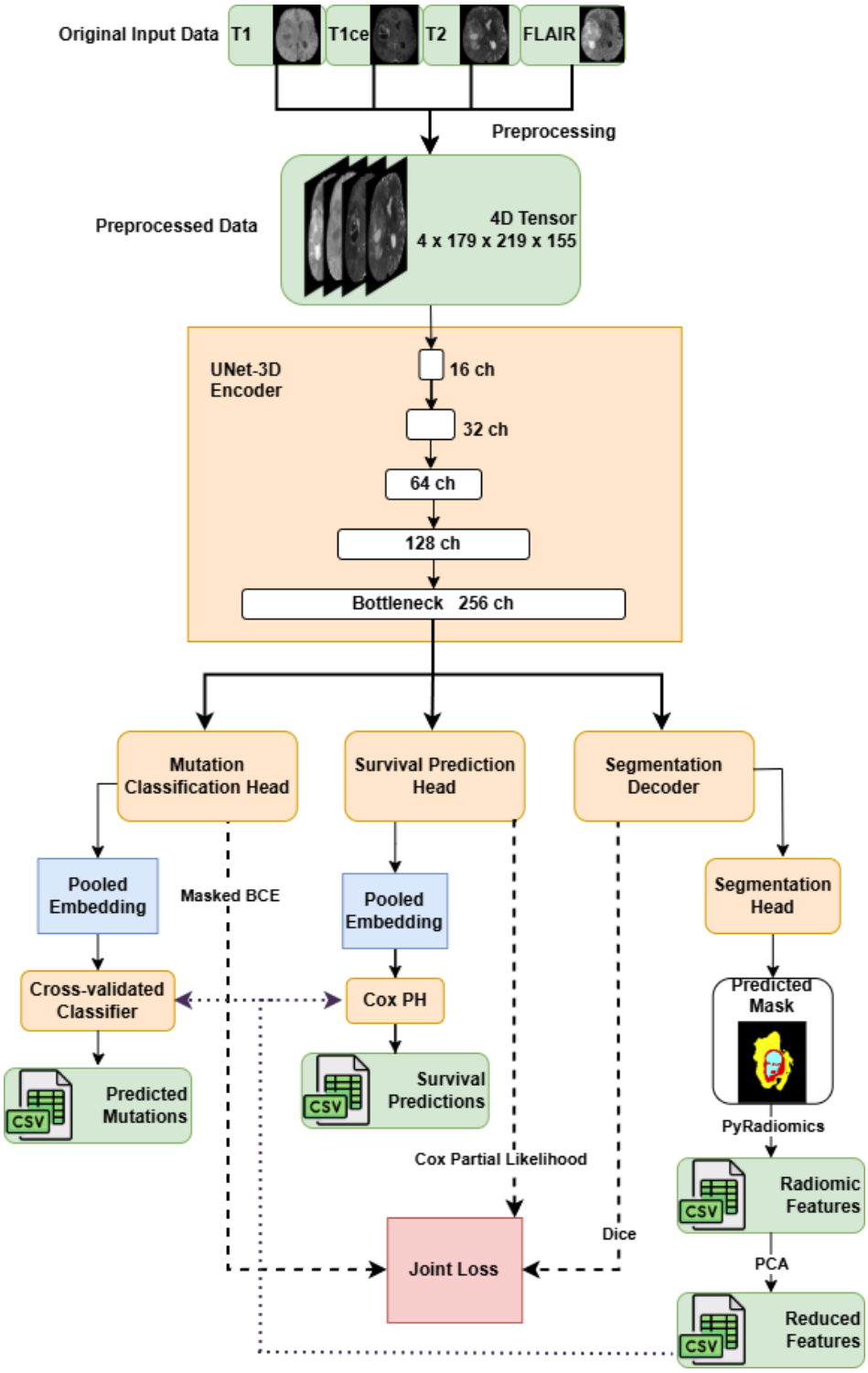

Figure 5: Multi-head model. A shared 3D U-Net encoder (feature depths 16→32→64→128 with a 256-channel bottleneck) feeds three task heads: segmentation, mutation, and survival. The dotted branch denotes the radiomics-fusion variant—experiments were run both with and without fusion of these PCA-compressed radiomic features into the mutation and survival heads.

Fig. 5 shows the unified, multi-head architecture. Four MRI sequences (T1, T1ce, T2, FLAIR) are preprocessed into a $4\times D\times H\times W$ tensor and passed through a shared 3D U-Net encoder.

The segmentation head applies a standard U-Net decoder to produce predicted masks. The mutation head globally pools the encoder's bottleneck features into a compact image embedding and produces per-gene mutation probabilities. The survival head similarly forms a global image embedding by pooling the encoder's bottleneck features, and concatenates the clinical covariates to output a scalar Cox risk score.

The dotted radiomics branch indicates the radiomics-fusion configuration. In this variant, the predicted segmentation masks are used to extract radiomic descriptors via PyRadiomics (see Sec. 3). These reduced radiomics features are then concatenated with the pooled image embeddings used in the mutation and survival heads and used for their corresponding predictions.

The shared encoder extracts a spatial feature map reused across segmentation, mutation, and survival prediction. Let the stacked MRI volumes be $x \in \mathbb{R}^{4\times D\times H\times W}$, and let g_θ denote the shared 3D U-Net encoder. The encoder transforms the inputs into a spatial representation

$$z = g_\theta(x), \tag{5}$$

where $z \in \mathbb{R}^{C\times D'\times H'\times W'}$ is the latent feature map, C is the channel depth at the bottleneck, and (D', H', W') are the encoder's spatial dimensions after downsampling.

The segmentation head decodes z to voxel-wise logits and probabilities,

$$U = h_{\text{seg}}(z), \qquad P = \text{softmax}(U), \tag{6}$$

with $U \in \mathbb{R}^{4\times D\times H\times W}$ and P the corresponding channel-wise softmax over the four classes. Here h_{seg} is the U-Net decoder. Additional details for h_{seg} appear in Section 5.3.

The mutation head forms a subject-level image embedding by global average pooling and produces per-gene logits and probabilities,

$$e = \text{pool}(z), \qquad a_j = h_{\text{mut},j}(e), \qquad p_j = \sigma(a_j) \tag{7}$$

where $e \in \mathbb{R}^C$ aggregates z over (D', H', W'), $h_{\text{mut},j}$ is the gene-specific output, $a_j \in \mathbb{R}$ is the binary logit, and $p_j \in (0, 1)$ is the corresponding sigmoid probability. Additional details for h_{mut} appear in Section 5.4.

The survival head similarly derives a global image representation and integrates clinical covariates to yield a scalar risk score,

$$r = \tilde{h}(z), \qquad \eta = h_{\text{surv}}(r, \text{clin}), \tag{8}$$

where r is a pooled embedding of z, clin denotes the encoded clinical covariates, and h_{surv} maps (r, clin) to the Cox proportional hazards risk $\eta \in \mathbb{R}$ used in the censored partial-likelihood objective [5]. Additional details for h_{surv} appear in Section 5.5.

In summary, the encoder learns a shared representation that supports three different types of supervision. This structure allows the model to share information efficiently across tasks while preserving task-specific detail where needed.

5.2.2 Multi-Task Optimization and Weighting

Parameters of the shared encoder and all task heads are optimized jointly using Adam. Let $\mathcal{T} = \{\text{seg}, \text{mut}, \text{surv}\}$ denote the task set, θ the shared-encoder parameters, and $\mathcal{L}_t$ the loss for task $t \in \mathcal{T}$. Gradients from all tasks contribute to the encoder update as a weighted sum:

$$\nabla_\theta \mathcal{L}_{\text{uw}} = \sum_{t\in\mathcal{T}} \alpha_t \, \nabla_\theta \mathcal{L}_t, \qquad \alpha_t = \tfrac{1}{2\sigma_t^2}, \tag{9}$$

where σ_t^2 is a learned variance for task t, and α_t is the weight that results from it.

To learn these weights from data rather than fixing them by hand, we follow homoscedastic uncertainty weighting [4], which augments each task loss with its variance

$$\mathcal{L}_{\text{uw}} = \sum_{t\in\{\text{seg,mut,surv}\}} \left(\frac{1}{2\sigma_t^2}\,\mathcal{L}_t + \frac{1}{2}\log\sigma_t^2 \right). \tag{10}$$

The term $1/(2\sigma_t^2)$ directly induces the effective weight α_t used in the gradient expression above, and the $\frac{1}{2}\log\sigma_t^2$ term prevents degenerate solutions by penalizing extreme σ_t^2 values.

The model learns the relative reliability of each task. In practice, mutation labels are sparse, so their learned variance becomes high and their weight is decreased, reducing their influence on the encoder. Segmentation labels, by contrast, are dense, so their variance becomes low and their weight is increased, giving segmentation more influence over the shared representation.

5.3 Segmentation Head

The segmentation head outputs four logits $U \in \mathbb{R}^{4\times D\times H\times W}$ (background and three tumor regions). For each spatial location (i, j, k), class probabilities are obtained by a channel-wise softmax:

$$P_c(i,j,k) = \frac{\exp(U_c(i,j,k))}{\sum_{m=1}^{4}\exp(U_m(i,j,k))}. \tag{11}$$

Training uses a multi-class soft Dice loss over the same four channels [7]. Dice focuses on the overlap of positive regions and avoids the misleadingly high scores that voxel accuracy can produce when background dominates. Let T be the one-hot encoding of ground truth and P the predicted probabilities:

$$\text{Dice}(c) = \frac{2\sum P_c T_c + \varepsilon}{\sum P_c + \sum T_c + \varepsilon}, \qquad L_{\text{Dice}} = 1 - \frac{1}{4}\sum_{c=1}^{4}\text{Dice}(c). \tag{12}$$

P_c and T_c denote the predicted and ground-truth masks for class c and ε is a stability constant.

At inference, discrete class labels are obtained by taking the channel-wise argmax of the logits:

$$\hat{Y}(b,i,j,k) = \arg\max_{c\in\{1,\dots,4\}} U(b,c,i,j,k). \tag{13}$$

Here U are the predicted logits, $\hat{Y}$ is the discrete label map, b indexes the subject in the batch, c indexes the class channel, and (i, j, k) index spatial coordinates.

During evaluation, predictions are combined into four standard BraTS composite regions—whole tumor (WT), enhancing tumor (ET), peritumoral edema (ED), and non-enhancing/necrotic core (NEC)—to assess segmentation quality. Region-wise Dice for each composite R is then

$$\text{Dice}(R) = \frac{2\,|P^R \cap G^R| + \varepsilon}{|P^R| + |G^R| + \varepsilon}, \qquad R \in \{\text{WT}, \text{ET}, \text{ED}, \text{NEC}\}, \tag{14}$$

where P^R and G^R are the predicted and ground-truth binary masks for region R.

5.4 Mutation Head

Each mutation head produces one logit z_j per gene j, which is converted to a probability using a sigmoid:

$$p_j = \frac{1}{1+e^{-z_j}}. \tag{15}$$

where $z_j \in \mathbb{R}$ is the logit for gene j and $p_j \in (0, 1)$ is the corresponding sigmoid probability.

Binary cross-entropy is used with a positive-class weight π_j to offset rare positives:

$$l_{\mathrm{WBCE}}(y_j, p_j;\, \pi_j) = -\big[\, \pi_j\, y_j \log(p_j) + (1 - y_j) \log(1 - p_j) \,\big], \qquad \pi_j = \frac{\#\mathrm{neg}_j}{\#\mathrm{pos}_j}. \tag{16}$$

Here, $y_j \in \{0, 1\}$ is the ground-truth label for gene j, π_j up-weights the positive class, and l_{WBCE} is the per-sample loss.

Because some genes have far fewer labeled subjects, each gene's loss is scaled by how under-represented it is. Let a_j be the number of labeled subjects for gene j, and let $\bar{a}$ be the geometric mean of $\{a_j\}$ across all genes. The per-gene weight is

$$w_j = \left(\frac{\bar{a}}{a_j}\right)^{1/2}. \tag{17}$$

Let $m_j \in \{0, 1\}$ indicate whether a label is available for gene j. Combining the above ideas, the overall objective becomes

$$L = \frac{1}{\sum_j m_j} \sum_j m_j\, w_j\, l_{\mathrm{WBCE}}(y_j, p_j; \pi_j). \tag{18}$$

At inference, each per-gene probability p_j is converted to a binary prediction using a threshold selected on the validation fold. We sweep thresholds $t \in [0, 1]$ and compute Youden's J statistic

$$J(t) = \mathrm{Sensitivity}(t) + \mathrm{Specificity}(t) - 1. \tag{19}$$

The threshold used at test time is

$$t^* = \arg\max_t J(t). \tag{20}$$

Gene-wise ROC–AUC is then reported across held-out folds to quantify overall discriminative performance.

5.5 Survival Head

The survival head predicts a continuous risk score $\eta = \beta^\top x$, where x denotes the fused imaging and clinical covariates and β are the learned coefficients. Training follows the Cox proportional hazards model, which estimates relative risk without assuming a specific survival-time distribution. The objective maximizes the Cox partial log-likelihood over n subjects:

$$\ell(\beta) = \sum_{i=1}^{n} \delta_i \left(\beta^\top x_i - \log \sum_{j \in \mathcal{R}(t_i)} \exp(\beta^\top x_j) \right), \tag{21}$$

where x_i is the covariate vector for subject i, t_i is the observed time (event or censoring), $\delta_i \in \{0, 1\}$ indicates an observed event, and $\mathcal{R}(t_i)$ is the risk set of subjects still under observation at time t_i. This formulation rewards higher predicted risk for patients who experience earlier events while correctly handling censored samples.

To prevent overfitting in a small cohort, we add an ℓ_2 regularization term (ridge penalty) [14] on the coefficients:

$$\mathcal{L}(\beta) = -\ell(\beta) + \lambda \|\beta\|_2^2, \tag{22}$$

In this expression, $\lambda \geq 0$ controls the amount of ℓ_2 shrinkage applied to the coefficient vector β.

Model performance is evaluated using two complementary discrimination metrics. The concordance index (C-index) [5] measures how well predicted risk scores preserve the true event-time ranking:

$$C = \frac{\sum_{i<j} \mathbf{1}[(t_i < t_j) \wedge (\eta_i > \eta_j)]}{\sum_{i<j} \mathbf{1}[t_i \neq t_j]}, \tag{23}$$

where η_i and η_j are predicted risk scores and $\mathbf{1}[\cdot]$ is an indicator function. A higher C indicates better concordance between predictions and outcomes.

In addition, time-dependent ROC–AUC is computed for a fixed clinical horizon T (e.g., median follow-up time) to quantify short-term discrimination—how well the model separates patients who experience an event before T from those who do not [5].

6 Experimental Results

We evaluate all three objectives—glioma segmentation, mutation classification, and survival analysis—using both single-head and multi-head models. The single-head networks act as baselines trained independently for each task. In contrast, the multi-head variants share a common encoder but branch into task-specific heads, allowing us to assess how shared representations affect performance across region-level and patient-level predictions.

6.1 Glioma Segmentation Results

We report Dice scores for the three BraTS subregions—NCR/NET, ED, and ET—by comparing binary region masks between prediction and ground truth, skipping subjects where the region is absent in both. Whole-tumor (WT) Dice is computed on the union of the three regions [7].

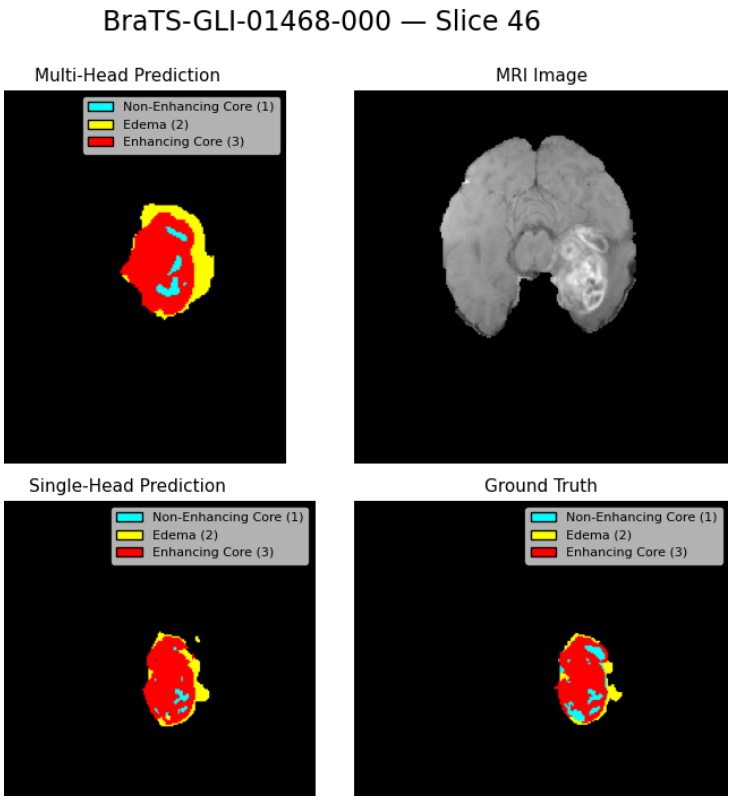

Figure 6: Segmentation Prediction Visual Comparison. Comparison of the ground-truth segmentation mask, the single-head predicted mask, and the multi-head predicted mask; the original T1CE slice is included for reference.

Figure 6 shows a case and slice with a clear distribution of all three BraTS regions, which allows a comprehensive comparison. In the ground truth and the single-head output, the edema forms a thin outline around the enhancing core. The multi-head model substantially enlarges this region, producing a much thicker edema ring. The enhancing core retains its oval shape in all outputs, though the single-head prediction follows the boundary more closely. For the

non-enhancing core, the multi-head model produces large disconnected blobs rather than the coherent structure seen in the ground truth, while the single-head model preserves the general shape but misses a few internal areas. Overall, the single-head prediction aligns better with the ground-truth geometry across regions.

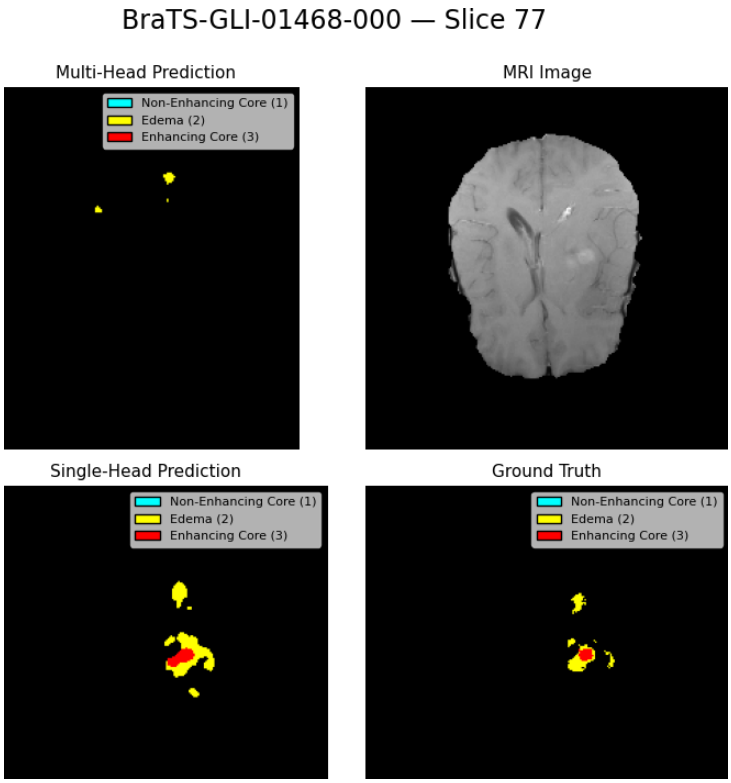

Figure 7: Segmentation Prediction Visual Comparison (different slice). Same subject as Figure 6, but a different axial slice. Ground truth, single-head prediction, and multi-head prediction are shown, with the original T1CE slice included for reference.

Figure 7 shows a sparser example where the ground truth contains several larger edema regions and a small enhancing core near the center. The multi-head model predicts only a few small regions, places them incorrectly, and misses the enhancing core entirely. The single-head model recovers the general pattern but slightly oversegments compared with the ground truth.

Table 1: Region-wise Dice (%) — single-head vs multi-head

Region	Single-Head	Multi-Head
Non-enhancing core	64.3	21.0
Edema	79.8	28.1
Enhancing core	79.7	25.4
Whole Tumor	92.1	55.0

Table 1 summarizes the quantitative gap between the two training methods. The single-head model substantially outperforms the multi-head model across all regions, with the largest deficits for the non-enhancing core and the enhancing core. These results match the qualitative patterns observed in Figures 6 and 7. The single-head model more consistently preserves the overall geometry and smaller structures of the tumor mask in comparison to the multi-head model.

The weaker performance of the multi-head model is consistent with the challenge of training one encoder to support very different objectives [4, 8]. Segmentation depends on detailed, boundary-focused features at voxel resolution [7], whereas mutation and survival provide only case-level supervision. When these objectives are optimized together, the shared encoder tends to shift toward broad, phenotype-related representations that benefit the patient-level tasks but reduce boundary accuracy for segmentation.

6.2 Mutation Classification Results

We examine the impact of several modeling choices on mutation classification. Our first comparison evaluates single-head models trained separately for each gene and tests whether incorporating copy-number alteration (CNA) labels improves mutation prediction. We then compare the single-head and multi-head results.

Table 2: Gene-wise ROC–AUC and Accuracy (single-head). Best-performing single-head configuration *(PCA on radiomics-only features)*, comparing mutation-only vs. mutation+CNA supervision. Accuracies use per-gene thresholds selected on validation via Youden's J.

Gene	AUC (Mut-only)	AUC (Mut+CNA)	Acc (Mut-only)	Acc (Mut+CNA)
EGFR	0.658	0.525	0.701	0.594
IDH1	0.595	0.511	0.509	0.353
PTEN	0.688	0.579	0.688	0.656
TP53	0.679	0.571	0.763	0.670
Avg	0.616	0.547	0.617	0.547

Table 2 shows that, in the single-head setting, using *mutation-only* supervision improves AUC and accuracy across all four genes compared with adding CNA labels. This likely reflects the greater noise and sparsity of CNA supervision relative to point mutations.

We next compare this mutation-only single-head baseline—a radiomics-only classifier with PCA—to two multi-head variants: one that uses shared features without radiomics, and one that incorporates radiomic features with the shared image embeddings [3, 1].

Table 3: Gene-wise ROC–AUC and Accuracy (multi-head). Single-head vs. multi-head (with/without radiomics fusion). Accuracies use per-gene validation thresholds (Youden's J).

Gene	AUC (Single)	AUC (Multi no rad)	AUC (Multi rad fusion)	Acc (Single)	Acc (Multi no rad)	Acc (Multi rad fusion)
EGFR	0.658	0.667	0.812	0.701	0.370	0.460
IDH1	0.595	0.708	0.604	0.509	0.820	0.820
PTEN	0.688	0.833	0.667	0.688	0.640	0.650
TP53	0.679	0.625	0.750	0.763	0.750	0.750
Avg	0.616	0.708	0.708	0.617	0.645	0.670

Table 3 shows that multi-head training improves average AUC compared with the single-head baseline (0.708 vs. 0.616), suggesting that shared representations enhance mutation classification overall. Accuracy follows a similar trend on average (0.645–0.670 vs. 0.617).

The impact of radiomic features differs across genes. On average, radiomics-fusion performs similarly to the no-radiomics multi-head variant in AUC (both 0.708) and shows a modest gain in accuracy (0.670 vs. 0.645). Gene-wise effects also vary: radiomics improves AUC for EGFR and TP53 but offers smaller or inconsistent changes for IDH1 and PTEN. These results indicate that the contribution of radiomic information is gene-dependent and does not benefit all targets uniformly.

6.3 Survival Analysis Results

Survival performance is evaluated on the test split using the concordance index (C-index) for time-to-event ranking and the ROC–AUC for event occurrence at horizon T (Section 4) [5]. We compare the single-head model, using its best-performing input configuration of PCA-reduced radiomic and deep features, with two jointly trained multi-head variants evaluated with and without radiomic features.

Table 4: Survival prediction (C-index and event AUC at T)

Input Data	C-index (time)	AUC (event at T)
Single-head (PCA on radiomics + deep)	0.724	0.917
Multi-head (no radiomics)	0.550	0.428
Multi-head (radiomics fusion)	0.500	0.483

Table 4 shows that the single-head pipeline performs best. Both multi-head variants fall well behind, and adding radiomics to the multi-head version does not recover the loss relative to the no-radiomics setting. This pattern is consistent with the earlier segmentation results: shared training pushes the encoder toward broad, phenotype-level features that support patient-level tasks but weaken boundary precision, producing poorer masks and, in turn, less reliable radiomic inputs for survival prediction [3]. Given the small sample size and the incomplete labels available for survival, joint optimization also introduces additional gradient competition in the shared encoder, leading to representations that are less effective for time-to-event ranking than those learned by a dedicated single-task model [8].

7 Limitations

This experiment has several limitations, reflecting both the difficulty of the underlying prediction tasks and the additional challenges introduced by multi-task training. First, joint optimization degraded segmentation performance: when segmentation, mutation, and survival are trained together, gradients from the case-level tasks compete with voxel-level supervision and push the shared encoder toward coarser, phenotype-oriented features. This reduces boundary accuracy and lowers region Dice (Table 1) [4, 8]. Second, supervision for the patient-level tasks is much weaker than for segmentation. The segmentation head is trained on dense voxel-wise masks for 1,251 subjects, whereas mutation prediction has labels for only 29 patients and survival only 91 patients. This gap in label density and sample size increases variance for the patient-level heads and makes multi-task balancing more fragile. Third, radiomics depends heavily on mask quality: weaker multi-head segmentations lead to noisier radiomic descriptors, which helps explain why late-fusion radiomics did not improve survival outcomes and benefited mutation prediction only selectively across genes.

8 Conclusion

This work implemented a unified three-head framework for glioma segmentation, mutation classification, and survival analysis from multimodal MRI. The single-task segmentation pipeline (3D U-Net with soft Dice) achieved strong region Dice, whereas the multi-head counterpart underperformed on all regions—a pattern consistent with gradient competition when voxel-level and case-level objectives share the same encoder. For mutation prediction, multi-head training improved the average AUC relative to a single-head image-embedding baseline, and radiomics late fusion provided additional, gene-specific gains. For survival, the best results came from a single-head, two-stage model using PCA on concatenated radiomics+deep features; the

multi-head variants did not match this performance, suggesting that stable mask-derived radiomics and task-specific optimization remain important for time-to-event modeling. Across tasks, PCA-based dimensionality reduction consistently stabilized training and improved performance relative to using raw high-dimensional features.

Because mutation and survival predictions rely on encoder embeddings or radiomics extracted from predicted masks, segmentation quality directly affects feature stability and downstream accuracy. Improving mask fidelity within the joint architecture—through architectural adjustments, better task balancing, or higher-capacity feature representations—should therefore raise performance for the patient-level heads. Two directions are especially promising: applying PCA to the encoder embeddings before mutation and survival prediction to reduce noise introduced by joint optimization, and increasing the encoder's bottleneck capacity (e.g., to 512 channels) so that the shared representation can better support three heterogeneous objectives. Together, these modifications may help narrow the performance gap between single- and multi-task pipelines.

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